# THE LCI-LOCUS HILBERT STACK COVERS THE BASE

THE STACKS PROJECT AUTHORS

*Editorial note.* No omission markers in-span; proof complete as written.

*Allowed standard prerequisites (editorial).*

- CM locus of flat lfp morphisms. more-morphisms-lemma-flat-finite-presentation-CM-open (cross-chapter, named). Existence of a Cohen-Macaulay point.
- Slicing. more-morphisms-lemma-slice-CM (cross-chapter, named). Produces the regular immersion Z -> V with Z = {v}.
- Unramified and lci permanence. spaces-morphisms-lemma-composition-unramified, spaces-morphisms-lemma-etale-unramified, spaces-morphisms-lemma-immersion-unramified, more-morphisms-lemma-regular-immersion-lci, spaces-morphisms-lemma-etale-smooth, spaces-morphisms-lemma-smooth-syntomic, spaces-more-morphisms-lemma-flat-lci, spaces-more-morphisms-lemma-composition-lci (cross-chapter, named). Certify the composed Z -> X_y is unramified lci.

*Source reference.* Original reference 00PJ: focused-seeds/geometry/S9-crit-lci-surjective/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 045U: focused-seeds/geometry/S9-crit-lci-surjective/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 069L: focused-seeds/geometry/S9-crit-lci-surjective/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 0570: focused-seeds/geometry/S9-crit-lci-surjective/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 06C8: focused-seeds/geometry/S9-crit-lci-surjective/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make

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*Source reference.* Original reference 06C9: focused-seeds/geometry/S9-crit-lci-surjective/reftags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 03XP: focused-seeds/geometry/S9-crit-lci-surjective/reftags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 03ZI: focused-seeds/geometry/S9-crit-lci-surjective/reftags.txt, source lines 12–12. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 04XX: focused-seeds/geometry/S9-crit-lci-surjective/reftags.txt, source lines 13–13. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 06CR: focused-seeds/geometry/S9-crit-lci-surjective/reftags.txt, source lines 14–14. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 06CQ: focused-seeds/geometry/S9-crit-lci-surjective/reftags.txt, source lines 15–15. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 06CP: focused-seeds/geometry/S9-crit-lci-surjective/reftags.txt, source lines 16–16. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $F : \mathcal{X} \to \mathcal{Y}$ be a 1-morphism of stacks in groupoids over $(Sch/S)_{fppf}$. Assume $F$ is representable by algebraic spaces, flat, surjective, and locally of finite presentation. Then*

$$\coprod_{d \geq 1} \mathcal{H}_{d,lci}(\mathcal{X}/\mathcal{Y}) \longrightarrow \mathcal{Y}$$

*is surjective on objects.*

*Proof.* It suffices to prove the following: For any field $k$ and object $y$ of $\mathcal{Y}$ over $\mathrm{Spec}(k)$ there exists an integer $d \geq 1$ and an object $(U, Z, y, x, \alpha)$ of $\mathcal{H}_{d,lci}(\mathcal{X}/\mathcal{Y})$ with $U = \mathrm{Spec}(k)$. Namely, in this case we see that $p$ is surjective on objects in the strong sense that an extension of the field is not needed.

Denote $X_y$ the algebraic space over $U = \mathrm{Spec}(k)$ representing the 2-fibre product $(Sch/U')_{fppf} \times_{y',\mathcal{Y},F} \mathcal{X}$. By assumption the morphism $X_y \to \mathrm{Spec}(k)$ is surjective and locally of finite presentation (and flat). In particular $X_y$ is nonempty. Choose a nonempty affine scheme $V$ and an étale morphism $V \to X_y$. Note that $V \to \mathrm{Spec}(k)$ is (flat), surjective, and locally of finite presentation (by Morphisms of Spaces, Definition 03XP). Pick a closed point $v \in V$ where $V \to \mathrm{Spec}(k)$ is Cohen-Macaulay (i.e., $V$ is Cohen-Macaulay at $v$), see More on Morphisms, Lemma 045U. Applying More on Morphisms, Lemma 0570 we find a regular immersion $Z \to V$ with $Z = \{v\}$. This implies $Z \to V$ is a closed immersion. Moreover, it follows that $Z \to \mathrm{Spec}(k)$ is finite (for example by Algebra, Lemma 00PJ). Hence $Z \to \mathrm{Spec}(k)$ is finite locally free of some degree $d$. Now $Z \to X_y$ is unramified as the composition of a closed immersion followed by an étale morphism (see Morphisms of Spaces, Lemmas 03ZI, 06CR, and 06CQ). Finally, $Z \to X_y$ is a local complete intersection morphism as a composition of a regular immersion of schemes and an étale morphism of algebraic spaces (see More on Morphisms, Lemma 069L and Morphisms of Spaces, Lemmas 04XX and 06CP and More on Morphisms of Spaces, Lemmas 06C9 and 06C8). The morphism $Z \to X_y$ corresponds to an object $x$ of $\mathcal{X}$ over $Z$ together with an isomorphism $\alpha : y|_Z \to F(x)$. We obtain an object $(U, Z, y, x, \alpha)$ of $\mathcal{H}_d(\mathcal{X}/\mathcal{Y})$. By what was said above about the morphism $Z \to X_y$ we see that it actually is an object of the subcategory $\mathcal{H}_{d,lci}(\mathcal{X}/\mathcal{Y})$ and we win. □